

Validity Analysis: LTZGLUE

Target context: Luxembourg Public Administration — Civil Servant NLU Deployment

Overall risk: **HIGH**

Deployment Context

Use case: In a public administration setting, a civil servant uses LLM-powered software to manage Luxembourgish citizen feedback by applying a model evaluated on topic classification, named entity recognition, intent detection, and sentiment analysis. The system automatically categorizes messages by domain and identifies key entities while classifying user intent and gauging sentiment to help detect frustrated users. These outputs enable the automated routing of requests to specific departments and the immediate prioritization of disgruntled or urgent correspondence, ensuring administrative actions are triggered accurately based on the content and tone of the message.

Target population: Civil servants in Luxembourgish government agencies who process digital correspondence and citizen inquiries written in the diverse orthographic styles of the Luxembourgish language.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	2	Significant gaps	high
Input Content 	2	Significant gaps	high
Input Form 	3	Moderate gaps	medium
Output Ontology 	1	Serious concern	high
Output Content 	2	Significant gaps	high
Output Form 	3	Moderate gaps	medium
Average	2.2		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

LTZGLUE is a credible first GLUE-style NLU benchmark for Luxembourgish and a valuable resource for general LTZ NLP capability assessment. However, for the specific deployment of routing citizen correspondence in Luxembourg public administration, validity is materially limited across the four HIGH-priority dimensions (IO, IC, OO, OC). The most severe gap is

output ontology: LTZGLUE produces single hard labels across taxonomies (RTL editorial topics, voice-assistant intents, ternary sentiment) that do not encode the administrative routing categories, multi-label structure, confidence scores, urgency scaling, or language-of-input axis the deployment requires. Input content reflects predominantly formal/institutional registers, with the authors themselves warning this does not represent informal and multilingual contexts — precisely the distribution of real citizen correspondence given Luxembourg's ~47% frontaliere workforce and entrenched code-switching. Annotation quality is moderate ($\kappa=0.45$ on sentiment) and lacks documented administrative-domain expertise. Input form and output form mismatches are moderate: text-to-label alignment is preserved, but length/language filtering and absence of calibration/multi-label metrics weaken external validity. The benchmark's authors explicitly caution against using performance as evidence of cultural or demographic coverage [Q121], and this caution applies directly here.

ID	Evidence
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Practical Guidance

What This Benchmark Measures

LTZGLUE measures general Luxembourgish NLU capability of encoder models and prompted LLMs across eight tasks (NER, topic, sentiment, acceptability, intent, RTE, headline acceptability). It is a useful instrument for selecting a base model with strong LTZ representation (e.g., MMBERT-base, LUXEMBERT, LTZ-E1) and for screening basic linguistic competence. Its NER component, in particular, can support entity extraction in citizen correspondence at a foundational level.

Construct Depth

Depth is shallow for the deployment's specific constructs. The benchmark probes language-level NLU (lexical, syntactic, basic semantic) with reasonable rigor, but it does not probe administrative-domain construct dimensions: routing topic taxonomy, state-vs-communal competency discrimination, frontaliere-specific terminology, formal-but-understated frustration detection, multi-label decisions, or language-of-input identification. Sentiment with $\kappa=0.45$ [Q27] and voice-assistant intents [Q154] cover only a thin slice of the operational construct space.

What Else You Need

(1) IO/OO: a Luxembourg-administrative topic taxonomy and multi-label dataset co-designed with civil servants, including state-vs-communal axis and confidence-threshold rules. (2) IC: a held-out evaluation set sampled from real citizen correspondence (anonymized under GDPR/CNPD guidance [WEB-9, WEB-10, WEB-11]) covering code-switched, orthographically variable, and frontaliere-specific content. (3) OC: re-annotation by administrative-experienced annotators with reported per-class agreement and a register-aware sentiment/urgency scheme. (4) OF: calibration evaluation (ECE, reliability) and multi-label metrics (micro/macro-F1, hamming loss). (5) Compliance: a DPIA and EU AI Act risk classification analysis [WEB-12, WEB-13] before deployment.

ID	Evidence
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)
WEB-9	https://www.luther-lawfirm.com/fileadmin/user_upload/Luxembourg_-_Data_Protection_Overview___Guidance_Note___DataGuidance.pdf
WEB-10	https://cnpd.public.lu/en.html
WEB-11	https://cnpd.public.lu/en/actualites/national/2024/09/rapport-annuel-2023.html
WEB-12	EU AI Act Annex III Category 5(a) high-risk classification potentially relevant — implies need for documented decision rules and oversight outputs not in benchmark (https://artificialintelligenceact.eu/annex/3/)
WEB-13	https://digital-strategy.ec.europa.eu/en/faqs/navigating-ai-act

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

LTZGLUE provides a GLUE-style suite of eight tasks covering binary and multi-class sentence- and token-level classification [Q11, Q67], which is a credible general-purpose NLU foundation for Luxembourgish. However, for the specific deployment — routing citizen correspondence to government departments — the taxonomies are misaligned. Topic classification covers only five RTL editorial categories (SPORTS, CULTURE, TECHNOLOGY, BUSINESS, ANIMALS) [Q48, Q153] with no administrative categories such as cross-border worker issues, housing, transport, tax, social security, or communal vs. national competency routing — all flagged as required in the elicitation and regional context. Intent detection labels are exclusively voice-assistant commands (alarm, reminder, weather) [Q154], with no administrative intents (complaint, document request, escalation). The benchmark contains no multi-label task formulation [Q67], whereas the deployment requires multi-label outputs [elicitation Q4]. The authors themselves acknowledge no unified benchmark currently exists for LTZ NLU [Q14]. Given IO is HIGH priority and the gaps span both omission (administrative categories) and irrelevance (voice-assistant intents) for this deployment, this constitutes significant taxonomic misalignment.

ID	Evidence
Q4	“Small and under-researched languages are particularly difficult to evaluate, as is the case with Luxembourgish (LTZ), the national language of Luxembourg, with around 400k speakers.” (p.1)
Q11	“In this section, we introduce the eight tasks for LTZGLUE. The set spans binary and multi-class sentence and token-level classification tasks. Together, these tasks cover a broad spectrum of linguistic and semantic phenomena and provide the first unified benchmark for evaluating LTZ NLP models.” (p.2)
Q14	“Yet progress remains uneven across tasks, and existing resources vary widely in size, domain, and annotation quality. No unified benchmark currently exists to evaluate LTZ language understanding consistently, a gap we aim to fill.” (p.2)
Q48	“From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS.” (p.4)
Q67	“Together, the eight tasks in LTZGLUE form a broad and balanced evaluation suite, covering four binary and four multi-class settings, sentence- and document-level inputs, as well as a token-level sequence-labelling task.” (p.5)
Q153	“topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology.” (p.15)
Q154	“slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm.” (p.15)

Strengths

- Provides eight diverse NLU task formulations including NER, topic classification, sentiment, RTE, and acceptability, establishing a baseline construct coverage for Luxembourgish [Q11, Q67]
- A substantial proportion of tasks are newly created for LTZ rather than direct translations, enabling some language-specific phenomena [Q69]
- NER schema includes PER, ORG, LOC, MISC, DATE [Q152] which has partial overlap with entities relevant to administrative correspondence (named persons, organizations, locations)

ID	Evidence
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Q11	"In this section, we introduce the eight tasks for LTZGLUE. The set spans binary and multi-class sentence and token-level classification tasks. Together, these tasks cover a broad spectrum of linguistic and semantic phenomena and provide the first unified benchmark for evaluating LTZ NLP models." (p.2)
Q67	"Together, the eight tasks in LTZGLUE form a broad and balanced evaluation suite, covering four binary and four multi-class settings, sentence- and document-level inputs, as well as a token-level sequence-labelling task." (p.5)
Q69	"In addition, a substantial proportion of the LTZGLUE tasks are newly created for LTZ rather than direct translations or simple repackaging, allowing the benchmark to reflect phenomena and usage patterns specific to the language." (p.5)
Q152	"ner: Perform Named Entity Recognition on the given sequence of sentence tokens. Output tags as lists of ner_tags. Allowed Tags: O, B-LOC, I-LOC, B-PER, I-PER, B-DATE, I-DATE, B-ORG, I-ORG, B-MISC, I-MISC." (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: The deployment requires categories for cross-border worker / frontaliere issues, housing/cost-of-living, transport, tax, social security, EU institutional matters, residency permits, communal services, public health, plus a state-vs-communal routing dimension and multi-label outputs with confidence scores [elicitation Q1, Q4; regional context administrative_domain_notes].

IO-2: Yes — none of LTZGLUE's topic classes (SPORTS, CULTURE, TECHNOLOGY, BUSINESS, ANIMALS) [Q48, Q153] correspond to administrative routing categories, and intent detection covers only voice-assistant commands [Q154]. No state-vs-communal competency dimension or multi-label formulation exists [Q67].

IO-3: Yes — the voice-assistant intent labels (alarm, reminder, weather sub-intents) [Q154] are construct-irrelevant for administrative correspondence routing. The RTE logical-entailment framing [Q59] is unrelated to routing decisions, though it may have ancillary value for general LLM evaluation.

IO-4: Major content-validity gap: the benchmark cannot assess routing taxonomy performance for cross-border worker issues, housing, transport, tax, communal vs. national competencies — the categories that drive operational value [regional context benchmark_validity_risks_summary.input_ontology_gap]. The absence of multi-label formulation [Q67] additionally prevents evaluation of the deployment's required output structure.

ID	Evidence
Q1	"This paper presents LTZGLUE, the first Natural Language Understanding (NLU) benchmark for Luxembourgish (LTZ) based on the popular GLUE benchmark for English." (p.1)
Q4	"Small and under-researched languages are particularly difficult to evaluate, as is the case with Luxembourgish (LTZ), the national language of Luxembourg, with around 400k speakers." (p.1)
Q14	"Yet progress remains uneven across tasks, and existing resources vary widely in size, domain, and annotation quality. No unified benchmark currently exists to evaluate LTZ language understanding consistently, a gap we aim to fill." (p.2)
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)
Q59	"Recognizing Textual Entailment (RTE) (Haim et al., 2006) is a classic NLU task featured in the original GLUE benchmark. Given a pair of texts A and B, the task consists of determining whether A is a logical premise of B." (p.5)
Q67	"Together, the eight tasks in LTZGLUE form a broad and balanced evaluation suite, covering four binary and four multi-class settings, sentence- and document-level inputs, as well as a token-level sequence-labelling task." (p.5)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)

Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)
WEB-2	OECD 2025: 50% of frontaliers from France, remainder split between Belgium and Germany — supports differentiated administrative-query profile (https://www.oecd.org/en/publications/2025/04/oecd-economic-surveys-luxembourg-2025_3eb782b5/full-report/reviving-productivity-growth_4509ab88.html)
WEB-5	~231,290 frontaliers (~47% of employment) — citizen-input distribution will be heavily multilingual (https://en.paperjam.lu/article/growth-in-number-of-cross-border)

Information Gaps

- Whether any LTZ-specific administrative-domain benchmark or taxonomy exists outside LTZGLUE — flagged as not searched in regional context
- Formal enumeration of communal vs. national competencies needed to define a routing taxonomy

Requires Expert Verification

- Final administrative topic taxonomy and multi-label structure must be co-designed with civil servants and legal/administrative experts

Remediation

Gap 1: No Luxembourg-administrative topic or intent taxonomy, no state-vs-communal routing axis, no multi-label formulation

Recommendation: Co-design an administrative topic taxonomy with national and communal civil servants covering at minimum: cross-border worker matters, housing/cost-of-living, transport, tax, social security, residency, communal services, EU institutional. Include a state-vs-communal routing dimension and a multi-label structure aligned with the deployment's actual case types.

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Benchmark inputs are drawn predominantly from RTL news articles, the Luxembourgish Online Dictionary, Wikipedia/Wikidata, and a translated English voice-assistant corpus [Q12, Q16, Q24, Q30, Q34, Q46, Q51]. The authors explicitly warn that 'most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts' [Q112], and that LTZ data 'often originate from a limited set of public domains' [Q119]. The deployment, however, expects orthographically variable, code-switched (Luxembourgish/French/German) citizen correspondence [elicitation Q2; regional context languages.code_switching_patterns]. Only the human-annotated NER sub-corpus from RTL user comments provides partial coverage of informal and code-mixed writing [Q41]. The intent detection corpus uses a voice-assistant register that has no LTZ reference, so terminology could not be systematically verified [Q55, Q56]. Given that ~47% of Luxembourg's workforce are frontaliers (mostly French-speaking) [WEB-5, WEB-2] and the language law requires the administration to respond in the citizen's chosen language [WEB-14], the benchmark's content distribution is materially narrower than the deployment input distribution. IC is HIGH priority and the documented gaps are substantial.

ID	Evidence
Q2	"Our tasks include common natural language processing tasks in binary and multi-class classification settings, including named entity recognition, topic classification, and intent classification." (p.1)
Q12	"Unless stated otherwise, the textual data used across most tasks stems from two main sources: (i)" (p.2)
Q16	"To construct this dataset, we use RTL news articles. We keep only documents from the twenty most frequent categories. We then filter articles by body length and title length, remove exact duplicate titles, randomly shuffle the remaining instances, and retain a fixed subset of 30k examples." (p.3)
Q24	"We use articles from RTL, randomly selected from the commentary and letter to the editor sections." (p.3)
Q30	"The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset." (p.3)
Q34	"The JUDGEWEL dataset (Plum et al., 2026) introduces an automatically constructed corpus for named entity recognition (NER) in LTZ, derived from Wikipedia and Wikidata." (p.4)
Q41	"It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE)." (p.4)
Q46	"To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories." (p.4)
Q51	"We constructed a new LTZ dataset for intent detection (ID) by translating the English xSID test and validation datasets (van der Goot et al., 2021)." (p.4)
Q55	"The main challenge in translating the English dataset stems from its register. The source segments consist of user commands for a voice-controlled AI assistant, representing a specialised spoken register for which there is no equivalent reference corpus in LTZ. This register is marked by domain-specific terminology and collocations (e.g., set an alarm, set a reminder, add to playlist), as well as non-standard spelling (e.g., all lowercase, missing punctuation)." (p.5)
Q56	"Due to the lack of LTZ references in this register, it was not possible to systematically verify the translated terminology." (p.5)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
Q119	"LTZ is a small language community, and linguistic data often originate from a limited set of public domains." (p.10)

WEB-2	OECD 2025: 50% of frontaliers from France, remainder split between Belgium and Germany — supports differentiated administrative-query profile (https://www.oecd.org/en/publications/2025/04/oecd-economic-surveys-luxembourg-2025_3eb782b5/full-report/reviving-productivity-growth_4509ab88.html)
WEB-5	~231,290 frontaliers (~47% of employment) — citizen-input distribution will be heavily multilingual (https://en.paperjam.lu/article/growth-in-number-of-cross-bord)
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark (https://data.legilux.public.lu/filestore/eli/etat/leg/loi/1984/02/24/n1/jo/fr/html/eli-etat-leg-loi-1984-02-24-n1-jo-fr.html)

Strengths

- Lothritz et al. NER sub-corpus from RTL online news comments provides partial informal and code-mixed writing coverage [Q40, Q41]
- Authors transparently document content limitations, register narrowness, and demographic-coverage caveats [Q110, Q112, Q119, Q121]
- Pre-training corpus for LTZ-E1 includes broader registers (user comments, webchat, transcribed podcasts and political speeches) [Q74], which may benefit downstream model representations even if task evaluation is narrower

ID	Evidence
Q40	"The NER dataset introduced by Lothritz et al. (2022), by contrast, is a fully human-annotated corpus derived from RTL online news comments." (p.4)
Q41	"It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE)." (p.4)
Q74	"The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024." (p.6)
Q110	"While LTZGLUE provides the first systematic benchmark for LTZ NLU, the dataset remains constrained by the availability and scope of existing resources. Several tasks rely on relatively small or domain-specific corpora, which limits the ecological validity of the results and restricts the range of linguistic phenomena covered." (p.9)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
Q119	"LTZ is a small language community, and linguistic data often originate from a limited set of public domains." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — citizen correspondence requires handling of orthographically variable Luxembourgish, French/German code-switching, frontier-specific terminology, and Luxembourg-specific institutional vocabulary [elicitation Q2; regional context languages.code_switching_patterns, cultural_norms_notes]. The benchmark inputs are dominated by RTL news and dictionary content [Q12, Q16, Q46], which do not encode this distribution.

IC-2: Partially. Sources are Luxembourgish in origin so cultural framing is locally grounded, but the institutional/journalistic register diverges from citizen-government correspondence [Q112]. The voice-assistant intent corpus is translated from English without LTZ register reference [Q55, Q56], introducing alien register material.

IC-3: Yes — the xSID-derived intent corpus (alarm/reminder/weather) [Q51, Q55] is Western voice-assistant content with no administrative analogue in Luxembourg; its inputs encode construct-irrelevant register. Authors note pre-training data overlap with LLMs may also inflate performance [Q111].

IC-4: INSUFFICIENT DOCUMENTATION — no Luxembourg civil-administration domain experts appear to have reviewed task inputs for representativeness; the elicitation flags this as needing real-time civil-servant feedback for progressive alignment [elicitation Q3].

IC-5: Documented content-validity issues: (a) formal/institutional register dominance [Q112], (b) absence of systematic code-switching coverage outside NER [Q41], (c) voice-assistant register import without LTZ reference [Q55, Q56], (d) no representation of frontier-specific administrative content [WEB-5].

ID	Evidence
Q2	“Our tasks include common natural language processing tasks in binary and multi-class classification settings, including named entity recognition, topic classification, and intent classification.” (p.1)
Q3	“We evaluate various pre-trained language models for LTZ to present an overview of the current capabilities of these models on the LTZ language.” (p.1)
Q12	“Unless stated otherwise, the textual data used across most tasks stems from two main sources: (i)” (p.2)
Q16	“To construct this dataset, we use RTL news articles. We keep only documents from the twenty most frequent categories. We then filter articles by body length and title length, remove exact duplicate titles, randomly shuffle the remaining instances, and retain a fixed subset of 30k examples.” (p.3)
Q41	“It covers a wider range of text types and registers, including informal and code-mixed writing, and focuses on four primary entity categories (PER, ORG, LOC, GPE).” (p.4)
Q46	“To construct the news topic classification (TC) dataset, we collected news articles from RTL, which provides content pre-assigned to editorial categories.” (p.4)
Q51	“We constructed a new LTZ dataset for intent detection (ID) by translating the English xSID test and validation datasets (van der Goot et al., 2021).” (p.4)
Q55	“The main challenge in translating the English dataset stems from its register. The source segments consist of user commands for a voice-controlled AI assistant, representing a specialised spoken register for which there is no equivalent reference corpus in LTZ. This register is marked by domain-specific terminology and collocations (e.g., set an alarm, set a reminder, add to playlist), as well as non-standard spelling (e.g., all lowercase, missing punctuation).” (p.5)
Q56	“Due to the lack of LTZ references in this register, it was not possible to systematically verify the translated terminology.” (p.5)
Q111	“In addition, some of the data sources used in this benchmark may already be included, in whole or in part, in the pre-training corpora of the large language models evaluated in this work. While the exact composition of proprietary pre-training datasets is typically not fully disclosed, this potential overlap cannot be entirely ruled out and may inflate performance estimates.” (p.9)
Q112	“Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts.” (p.9)
WEB-5	~231,290 frontaliers (~47% of employment) — citizen-input distribution will be heavily multilingual (https://en.paperjam.lu/article/growth-in-number-of-cross-bord)
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark (https://data.legilux.public.lu/filestore/eli/etat/leg/loi/1984/02/24/n1/jo/fr/html/eli-etat-leg-loi-1984-02-24-n1-jo-fr-html.html)

Information Gaps

- No quantification of code-switching frequency in any task split
- No measure of orthographic-variation coverage in task inputs
- Whether RTL user-comment NER sub-corpus contains enough French/German switching to represent deployment distribution

Requires Expert Verification

- Sampling and qualitative review of LTZGLUE task inputs by civil-administration linguists to confirm representativeness for citizen-correspondence content
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Remediation

Gap 1: Benchmark inputs are dominated by formal/institutional registers and filter out non-Luxembourgish text; code-switching and orthographic variation are under-represented

Recommendation: Construct an internal evaluation corpus from anonymized real citizen correspondence (with CNPD-compliant DPIA) that explicitly preserves French/German code-switching, frontier terminology, and orthographic variation. Use it as an out-of-distribution stress test alongside LTZGLUE.

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Form mismatch is moderate. Both benchmark and deployment are text-only in Latin script, so there is no modality or script-direction violation [regional context infrastructure_notes.deployment_modality; writing_systems]. Tokenization (BPE, vocab 50,368, max sequence 1,024) [Q128, Q129] is suitable for typical citizen-correspondence lengths (40–400 words covers many but not all messages [regional context infrastructure_notes.expected_input_length, Q47]). However, signal-distribution mismatch persists at the form level: task inputs are filtered to remove non-Luxembourgish text via OpenLID and constrained to 40–400 words [Q47], which removes precisely the multilingual and length-extreme inputs the deployment will encounter. The benchmark's text register (formal news, dictionary examples, transcribed political speeches) [Q74, Q16, Q30] differs from informal, orthographically variable citizen messages. IF is MODERATE priority; the form-level alignment is partial because text-text matching is achieved, but distributional filtering and register narrowness reduce external validity.

ID	Evidence
Q16	"To construct this dataset, we use RTL news articles. We keep only documents from the twenty most frequent categories. We then filter articles by body length and title length, remove exact duplicate titles, randomly shuffle the remaining instances, and retain a fixed subset of 30k examples." (p.3)
Q30	"The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset." (p.3)
Q47	"We applied a series of preprocessing steps to ensure data quality. Specifically, we removed articles identified as non-Luxembourgish by OpenLID (Burchell et al., 2023), as well as those containing fewer than 40 words or more than 400 words." (p.4)
Q74	"The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024." (p.6)
Q128	"Both models share a GPTNeoXTokenizerFast tokenizer (Black et al., 2022), a BPE-based tokenizer, which we train on the entire pre-training set, using a minimum frequency of two and a vocabulary size of 50,368." (p.12)
Q129	"We use a constant batch size of 1024 packed sequences, where both models have a max sequence length of 1024." (p.12)

Strengths

- Text-only modality matches deployment exactly [regional context infrastructure_notes.deployment_modality]
- Latin-script writing system aligns with all three administrative languages [regional context writing_systems]
- BPE tokenizer trained on broad LTZ pre-training corpus including comments, webchat, podcasts may handle orthographic variation reasonably [Q74, Q128]
- Sequence length cap of 1,024 tokens is sufficient for most short-to-medium correspondence

ID	Evidence
Q74	"The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024." (p.6)

Q128	“Both models share a GPTNeoXTokenizerFast tokenizer (Black et al., 2022), a BPE-based tokenizer, which we train on the entire pre-training set, using a minimum frequency of two and a vocabulary size of 50,368.” (p.12)
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

- IF-1:** Both source and target are text. Length filter (40–400 words) [Q47] excludes very short and very long messages that are realistic in administrative correspondence (e.g., one-line inquiries, multi-page complaints).
- IF-2:** Yes — Luxembourg has 99.0% internet penetration [WEB-7] and advanced digital infrastructure [regional context infrastructure_notes.connectivity]; data-capture specifications match.
- IF-3:** Use-case-specific differences: filtering of non-Luxembourgish text via OpenLID [Q47] removes code-switched inputs that the deployment must actually process. Pre-training data is broader [Q74] but downstream task data is narrower in register.
- IF-4:** External-validity concerns from (a) length filtering excluding tail of distribution, (b) language-detection filtering removing multilingual inputs, (c) register narrowness in downstream task data [Q47, Q112].

ID	Evidence
Q47	“We applied a series of preprocessing steps to ensure data quality. Specifically, we removed articles identified as non-Luxembourgish by OpenLID (Burchell et al., 2023), as well as those containing fewer than 40 words or more than 400 words.” (p.4)
Q74	“The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024.” (p.6)
Q112	“Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts.” (p.9)
Q129	“We use a constant batch size of 1024 packed sequences, where both models have a max sequence length of 1024.” (p.12)
WEB-7	99.0% internet penetration — confirms text-output channel is universally accessible (https://datareportal.com/reports/digital-2024-luxembourg)

Information Gaps

- Empirical length distribution of expected citizen correspondence
- Empirical proportion of code-switched messages in real digital channels

Requires Expert Verification

- Sample-level review of incoming citizen messages to validate that benchmark form constraints (length, language-purity filtering) do not exclude operationally important inputs

Remediation

Gap 1: Length filter (40–400 words) and language-purity filter (OpenLID) exclude tail-distribution inputs the deployment will face

Recommendation: Construct a supplementary evaluation slice without length and language-purity filters to characterize model behavior on very short, very long, and multilingual inputs; report performance separately on these slices.

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Severe structural misalignment between benchmark output ontology and deployment requirements. All eight LTZGLUE tasks produce single hard labels [\[Q67\]](#) across narrow label spaces: binary headline acceptability [\[Q148\]](#), three-class sentiment [\[Q149\]](#), binary/four-class linguistic acceptability [\[Q33, Q151\]](#), BIO-tagged NER over PER/ORG/LOC/MISC/DATE [\[Q44, Q152\]](#), five-class topic [\[Q48, Q153\]](#), nine voice-assistant intents [\[Q154\]](#), and binary entailment [\[Q155\]](#). The deployment explicitly requires multi-label classification with confidence scores for routing and human-fallback escalation [\[elicitation Q4\]](#), and demands administrative topic categories (cross-border worker, housing, transport, tax, communal vs. national) and urgency/frustration scales — none of which appear in any task. Sentiment's three-class scheme [\[Q149\]](#) cannot encode urgency or formal-but-understated frustration cues that are operationally significant per the elicitation [\[elicitation Q3\]](#) and regional cultural notes. EU AI Act considerations [\[WEB-12\]](#) and the language law's requirement to detect citizen language choice [\[WEB-14\]](#) introduce further required output dimensions (language-of-input, possibly risk flags) absent from the benchmark. OO is HIGH priority. This is the most severe dimension violation.

ID	Evidence
Q3	"We evaluate various pre-trained language models for LTZ to present an overview of the current capabilities of these models on the LTZ language." (p.1)
Q4	"Small and under-researched languages are particularly difficult to evaluate, as is the case with Luxembourgish (LTZ), the national language of Luxembourg, with around 400k speakers." (p.1)
Q33	"The binary dataset distinguishes between correct (1) and incorrect (0), for which the label 0 encompasses the categories Verb, Adj, Syntax and Ortho." (p.4)
Q44	"To ensure compatibility, the tag set is harmonised by merging the GPE and LOC categories into a single location label, while retaining PER, ORG, and MISC unchanged." (p.4)
Q48	"From the available categories, we focused on five principal domains: SPORTS, CULTURE, TECHNOLOGY, BUSINESS, and ANIMALS." (p.4)
Q67	"Together, the eight tasks in LTZGLUE form a broad and balanced evaluation suite, covering four binary and four multi-class settings, sentence- and document-level inputs, as well as a token-level sequence-labelling task." (p.5)
Q148	"headline_classification: Decide if the given title/headline fits the text. Output True or False." (p.15)
Q149	"sentiment_analysis: Classify sentiment of the text. Allowed labels: positive, neutral, negative." (p.15)
Q151	"linguistic_acceptability_multilabel: Detect if the sentence is correct or if some element is wrong. If the sentence is correct, Output: correct. If it is not, Output the label referencing the wrong element: syntax, verb, ortho or adj." (p.15)
Q152	"ner: Perform Named Entity Recognition on the given sequence of sentence tokens. Output tags as lists of ner_tags. Allowed Tags: O, B-LOC, I-LOC, B-PER, I-PER, B-DATE, I-DATE, B-ORG, I-ORG, B-MISC, I-MISC." (p.15)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)
Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)
Q155	"recognizing_textual_entailment: Determine if the information in the second sentence is entailed in the first one. Output: 0 or 1." (p.15)

WEB-12	EU AI Act Annex III Category 5(a) high-risk classification potentially relevant — implies need for documented decision rules and oversight outputs not in benchmark (https://artificialintelligenceact.eu/annex/3/)
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark (https://data.legilux.public.lu/filestore/eli/etat/leg/loi/1984/02/24/n1/jo/fr/html/eli-etat-leg-loi-1984-02-24-n1-jo-fr.html.html)

Strengths

- Standard, well-documented label schemas (BIO for NER, ternary sentiment) that are interoperable with downstream pipelines [Q44, Q149, Q152]
- NER label set (PER, ORG, LOC, MISC, DATE) is partially useful for entity extraction within administrative messages [Q152]

ID	Evidence
Q44	"To ensure compatibility, the tag set is harmonised by merging the GPE and LOC categories into a single location label, while retaining PER, ORG, and MISC unchanged." (p.4)
Q149	"sentiment_analysis: Classify sentiment of the text. Allowed labels: positive, neutral, negative." (p.15)
Q152	"ner: Perform Named Entity Recognition on the given sequence of sentence tokens. Output tags as lists of ner_tags. Allowed Tags: O, B-LOC, I-LOC, B-PER, I-PER, B-DATE, I-DATE, B-ORG, I-ORG, B-MISC, I-MISC." (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels are designed for general NLU evaluation, not administrative routing. Topic classes (sports, culture, technology, business, animals) [Q153] have no overlap with required routing categories.

OO-2: Missing categories include: cross-border worker / frontaliere issues, housing/cost-of-living, transport, tax, social security, EU institutional matters, residency permits, communal services, public health, plus a state-vs-communal routing axis and an urgency/escalation axis [regional context administrative_domain_notes; elicitation Q1, Q4].

OO-3: The voice-assistant intent labels (alarm/reminder/weather) [Q154] encode a non-administrative, non-Luxembourg-public-sector value framing — they are operational categories from a consumer assistant context, not civic communication.

OO-4: Stakeholder-driven taxonomy redesign is necessary; the elicitation confirms that routing logic requires multi-label outputs and confidence-based escalation [elicitation Q4], implying co-design with civil servants and legal experts.

OO-5: Structural validity is violated: the benchmark's single-label space cannot represent the multi-label decision structure [Q67 vs. elicitation Q4]. Content validity is violated through the absence of required administrative categories. External validity is at risk because performance on these tasks does not predict performance on the operational decision structure.

ID	Evidence
Q1	"This paper presents LTZGLUE, the first Natural Language Understanding (NLU) benchmark for Luxembourgish (LTZ) based on the popular GLUE benchmark for English." (p.1)
Q4	"Small and under-researched languages are particularly difficult to evaluate, as is the case with Luxembourgish (LTZ), the national language of Luxembourg, with around 400k speakers." (p.1)

Q67	"Together, the eight tasks in LTZGLUE form a broad and balanced evaluation suite, covering four binary and four multi-class settings, sentence- and document-level inputs, as well as a token-level sequence-labelling task." (p.5)
Q149	"sentiment_analysis: Classify sentiment of the text. Allowed labels: positive, neutral, negative." (p.15)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)
Q154	"slot_intent_detection: Detect the intent for the text given. Allowed intents: reminder/show_reminders, weather/find, reminder/set_reminder, reminder/cancel_reminder, alarm/snooze_alarm, alarm/show_alarms, alarm/set_alarm, nalarm/cancel_alarm, nalarm/time_left_on_alarm." (p.15)
WEB-12	EU AI Act Annex III Category 5(a) high-risk classification potentially relevant — implies need for documented decision rules and oversight outputs not in benchmark (https://artificialintelligenceact.eu/annex/3/)
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark (https://data.legilux.public.lu/filestore/eli/etat/leg/loi/1984/02/24/n1/jo/fr/html/eli-etat-leg-loi-1984-02-24-n1-jo-fr-html.html)

Information Gaps

- Final list of required routing categories and confidence-threshold rules

Requires Expert Verification

- Co-design of administrative topic taxonomy and multi-label/confidence scheme with civil servants, legal counsel (CNPD), and AI Act compliance officers

Remediation

Gap 1: All tasks emit single hard labels with narrow non-administrative taxonomies; no confidence scores, no multi-label outputs, no urgency scale, no language-of-input output

Recommendation: Define a deployment-specific output schema with multi-label routing categories, calibrated confidence scores tied to escalation thresholds, an urgency scale validated against civil-servant judgments, and a language-of-input field to satisfy the 1984 language law obligation [WEB-14].

ID	Evidence
WEB-14	Language law requires response in citizen's chosen language — implies language-of-input output dimension absent from benchmark (https://data.legilux.public.lu/filestore/eli/etat/leg/loi/1984/02/24/n1/jo/fr/html/eli-etat-leg-loi-1984-02-24-n1-jo-fr-html.html)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation provenance and quality are mixed and weakly tied to civil-administration norms. Sentiment was annotated by two native LTZ speakers with Cohen's Kappa 0.45 — moderate agreement — and disagreements resolved by consensus [Q25, Q26, Q27, Q28]; no documentation of administrative-register expertise. JUDGEWEL NER relies on LLM-as-judge with 'minimal human verification' [Q37]. RTE used ChatGPT-5-mini to verify labels, finding nearly 10% incorrect labels [Q63, Q64], with manual correction afterwards [Q65]. The intent detection translation lacked LTZ register references and could not be systematically verified [Q56]. Authors explicitly caution against 'using benchmark performance as evidence of cultural or demographic coverage' [Q121] and note 'models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices' [Q120]. For a deployment where sentiment must capture formal-but-understated frustration in civil correspondence [elicitation Q3, regional context cultural_norms_notes], $\kappa=0.45$ with consumer-facing news comments as source material is insufficient. OC is HIGH priority.

ID	Evidence
Q3	"We evaluate various pre-trained language models for LTZ to present an overview of the current capabilities of these models on the LTZ language." (p.1)
Q25	"In total, we extract 4,583 sentences, which are then annotated by two native speakers of LTZ." (p.3)
Q26	"Annotators are instructed to label each sentence, and to use unsure only when they would otherwise randomly use the other labels." (p.3)
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q28	"For the final set, the annotators agree on a label in cases of label disagreement." (p.3)
Q37	"The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds." (p.4)
Q56	"Due to the lack of LTZ references in this register, it was not possible to systematically verify the translated terminology." (p.5)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q65	"We found that the quality improvement step often corrected intentional logical contradictions or factual inaccuracies rather than keeping the original semantics. We therefore adjusted the sentences manually such that they corresponded to the ground truth again, while keeping false positives intact." (p.5)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Strengths

- Native LTZ speakers conducted sentiment annotation [Q25] and intent translation [Q52], grounding labels in language-native judgment
- Lothritz et al. NER sub-corpus is fully human-annotated with 'high-precision' result [Q42]
- RTE underwent multi-stage label verification with manual correction [Q63, Q64, Q65]

- Authors transparently disclose annotation limitations [Q113, Q120, Q121]

ID	Evidence
Q25	"In total, we extract 4,583 sentences, which are then annotated by two native speakers of LTZ." (p.3)
Q42	"Annotation was conducted manually, yielding a smaller but high-precision dataset." (p.4)
Q52	"The translations were performed by an LTZ native speaker. In cases of uncertainty, additional native LTZ speakers were consulted." (p.4)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q65	"We found that the quality improvement step often corrected intentional logical contradictions or factual inaccuracies rather than keeping the original semantics. We therefore adjusted the sentences manually such that they corresponded to the ground truth again, while keeping false positives intact." (p.5)
Q113	"Although we draw on established GLUE-style tasks, some annotation decisions and class distributions are necessarily influenced by resource constraints. Certain tasks exhibit label imbalance or rely on automatic preprocessing, which may introduce biases that we cannot fully quantify." (p.9)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: INSUFFICIENT DOCUMENTATION on whether labels reflect civil-administration stakeholder perspectives. Sentiment annotators are described only as native LTZ speakers [Q25]; no administrative or civil-service background is documented.

OC-2: Likely disagreement risk is high for sentiment/urgency: regional culture features formal, understated expression of frustration [regional context cultural_norms_notes]; news-comment-trained annotators with $\kappa=0.45$ [Q27] are unlikely to encode that register reliably.

OC-3: Documentation is partial — annotator counts and native-speaker status reported [Q25, Q52], but no demographic, professional, or domain-expertise breakdown provided. No Datasheet/Data Statement is referenced.

OC-4: Re-annotation by civil-servant or administration-experienced annotators is recommended; the elicitation already provisions a real-time civil-servant feedback loop [elicitation Q3] but pre-deployment re-annotation would be additionally valuable.

OC-5: Sentiment disagreements were resolved by consensus between two annotators [Q28], which can erase minority perspectives; this is a risk for a population with diverse linguistic backgrounds (frontaliers, non-native LTZ writers).

OC-6: Convergent validity is at risk because labels may not correlate with civil-servant judgments of administrative correspondence; external validity is at risk because the source distribution (RTL comments, dictionary, voice-assistant) differs from citizen-administration correspondence [Q112, Q119].

ID	Evidence
Q3	"We evaluate various pre-trained language models for LTZ to present an overview of the current capabilities of these models on the LTZ language." (p.1)
Q25	"In total, we extract 4,583 sentences, which are then annotated by two native speakers of LTZ." (p.3)
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q28	"For the final set, the annotators agree on a label in cases of label disagreement." (p.3)
Q52	"The translations were performed by an LTZ native speaker. In cases of uncertainty, additional native LTZ speakers were consulted." (p.4)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
Q119	"LTZ is a small language community, and linguistic data often originate from a limited set of public domains." (p.10)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Information Gaps

- Annotator demographics, professional background, civil-administration experience
- Whether sentiment label boundaries were validated against civil-correspondence-specific examples
- Granular agreement statistics by class (e.g., neutral vs. negative) to assess understated-frustration handling

Requires Expert Verification

- Civil-servant review of a held-out sample of sentiment labels for civil-correspondence relevance
- Re-annotation of a small evaluation set by administrative-domain annotators to estimate convergent validity

Remediation

Gap 1: Sentiment annotators lack documented administrative-register expertise; $\kappa=0.45$ is moderate; LLM-assisted labeling introduces uncertainty in RTE and JUDGEWEL

Recommendation: Re-annotate a calibration sample of sentiment and any new routing categories with civil servants or administration-experienced annotators; report per-class agreement and disagreement patterns; use the deployment's human-in-the-loop feedback to drive ongoing label refinement.

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is text-classification labels evaluated by macro-F1 with results averaged over three runs and standard deviations reported [Q93, Q96]. This is a standard, well-documented evaluation methodology and matches the deployment's text-to-label format at surface level. However, the deployment requires confidence scores for routing thresholds and multi-label outputs for messages spanning multiple departments [elicitation Q4]. LTZGLUE reports only macro-F1 with no calibration metrics (e.g., ECE, Brier score), no per-class precision/recall thresholds, and no multi-label evaluation methodology. Invalid LLM outputs are simply discarded [Q90, Q91], which masks the operational reality that malformed outputs in deployment must be handled (escalated). OF is MODERATE priority. The form is text-to-text-label as required, but the metric set and label cardinality do not match the deployment's operational form.

ID	Evidence
Q4	"Small and under-researched languages are particularly difficult to evaluate, as is the case with Luxembourgish (LTZ), the national language of Luxembourg, with around 400k speakers." (p.1)
Q90	"Prompted LLMs do not always produce well-formed outputs and may return an incorrect number of predictions for a given task; such outputs are discarded prior to evaluation." (p.7)
Q91	"All reported scores are computed on the remaining valid predictions per model." (p.7)
Q93	"Table 6 shows F1 scores for all models across all tasks (see Appendix 7.9 for full results)." (p.7)
Q96	"Table 6: Test F1 scores across all ItzGLUE tasks. Encoder results are averaged over three runs with standard deviations as subscripts. Prompted LLMs were evaluated once; we report macro-F1 only." (p.8)

Strengths

- Text-output modality matches deployment [Q93]
- Multi-run evaluation with standard deviations reported for encoder models [Q89, Q96] supports meaningful comparison
- Hyperparameter search via Bayesian optimization with early stopping is methodologically sound [Q136, Q138]
- Class-balanced loss for imbalanced tasks [Q92] is a conscientious design choice

ID	Evidence
Q89	"For encoder-based models, results are reported as averages over multiple runs (see Appendix 7.2 for more details)." (p.7)
Q92	"For the supervised models, since the linguistic acceptability and sentiment analysis datasets are highly imbalanced, when fine-tuning on these tasks we use class-balanced loss based on effective size (Cui et al., 2019) with a beta of 0.99." (p.7)
Q93	"Table 6 shows F1 scores for all models across all tasks (see Appendix 7.9 for full results)." (p.7)
Q96	"Table 6: Test F1 scores across all ItzGLUE tasks. Encoder results are averaged over three runs with standard deviations as subscripts. Prompted LLMs were evaluated once; we report macro-F1 only." (p.8)
Q136	"Though we do not aim to optimise performance in our evaluation, we conduct basic hyperparameter sweeps for each model and task combination in order to provide a fairer comparison across models." (p.13)
Q138	"In order to reduce the computational demand of the sweeps, we use Bayesian search with early stopping after three iterations, and cap each sweep at 30 runs, for 1,440 total runs across all models and tasks (and an additional 96 to finetune the two additional seeds)." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Text output matches deployment, but the deployment additionally requires confidence scores and multi-label outputs [elicit¹ation Q4], which the benchmark’s macro-F1 single-label evaluation does not assess.

OF-2: Not applicable — deployment is text-only [regional context infrastructure_notes.deployment_modality], and benchmark is text-only [Q93].

OF-3: Civil servants are professionally literate [regional context target_population.civil_servant_profile]; literacy is not a constraint. Citizens have 99.0% internet penetration [WEB-7]. No accessibility-driven output modality concerns.

OF-4: External-validity gap: macro-F1 alone cannot validate the calibration and multi-label behaviors required for routing escalation logic; invalid outputs being discarded [Q90, Q91] hides a deployment-relevant failure mode.

ID	Evidence
Q4	“Small and under-researched languages are particularly difficult to evaluate, as is the case with Luxembourgish (LTZ), the national language of Luxembourg, with around 400k speakers.” (p.1)
Q90	“Prompted LLMs do not always produce well-formed outputs and may return an incorrect number of predictions for a given task; such outputs are discarded prior to evaluation.” (p.7)
Q91	“All reported scores are computed on the remaining valid predictions per model.” (p.7)
Q93	“Table 6 shows F1 scores for all models across all tasks (see Appendix 7.9 for full results).” (p.7)
Q96	“Table 6: Test F1 scores across all ItzGLUE tasks. Encoder results are averaged over three runs with standard deviations as subscripts. Prompted LLMs were evaluated once; we report macro-F1 only.” (p.8)
WEB-7	99.0% internet penetration — confirms text-output channel is universally accessible (https://datareportal.com/reports/digital-2024-luxembourg)

Information Gaps

- No calibration metrics (ECE, reliability diagrams) reported
- No multi-label evaluation methodology defined
- No analysis of malformed-output rate as an operational signal

Requires Expert Verification

- Specification of operational confidence thresholds and escalation-rate targets by deployment owners

Remediation

Gap 1: Macro-F1 only; no calibration metrics; invalid outputs are silently discarded; no multi-label evaluation

Recommendation: Add calibration metrics (ECE, reliability diagrams), multi-label metrics (micro/macro-F1, hamming loss, subset accuracy), and treat malformed-output rate as an explicit deployment KPI rather than discarding such cases.